655.

On Products of the Slow Combustion of Ethyl Ether.

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I have already reported on a new product of the slow combustion of ether[1)]; I obtained it crystallized in rhombic prisms by rapid evaporation of the liquid formed during this combustion process. To prepare the latter I used the apparatus drawn in the figure below,

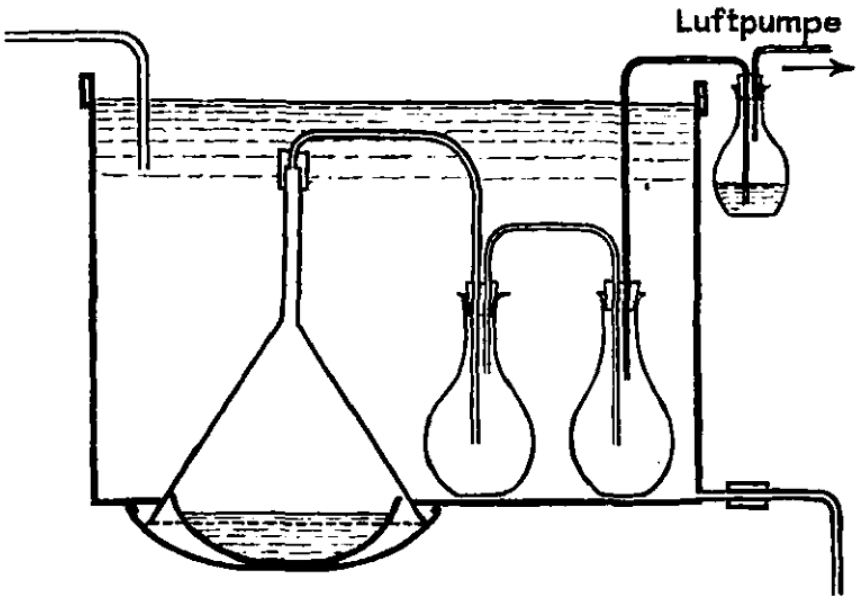


Fig. 1. Apparatus for preparing the starting liquid under cooling water (connected to the air pump, "Luftpumpe").

whose arrangement differs from the one described earlier in that the preparation of the starting liquid takes place entirely under cooling water.

The ether is placed in porcelain dishes, always set two inside one another; the combustion process is initiated by a faintly glowing sheet of platinum set upright in the ether dish and maintained by supply of air. The decomposition products partly run down the walls of the overlying funnel into the outer dish, and partly condense in the receivers into the acidic liquid smelling of aldehyde, the so-called **lamp acid**.

A tin pot serves as the cooling vessel; the funnel is cemented into its interior through a cutout in the bottom. The air current led through the apparatus enters between the dishes and the funnel and finally passes the small flask located outside, containing a little water, which mediates the regulation of the air flow.

The crystals obtained by evaporating the collected liquid over sulfuric acid gave on average the empirical composition $C_{11}H_{33}O_{21}$. Later I took the simplest form of this substance to be $C_2H_6O_4$[2)], on the basis of quantitative determinations of the decomposition products that appear, in particular:

— formic acid, methyl aldehyde (formaldehyde), hydrogen on the action of alkalis,

— formic acid, hexamethylenamine, oxygen on the action of ammonia.

At the same time the decomposition products obtained partly indicated a substance containing the methyl aldehyde group as its core, and partly other characteristic reactions pointed to a compound endowed with active oxygen, similar to hydrogen peroxide.

But I obtained closer information about its constitution only through a nitrogen-containing derivative, the

Hexaoxymethylenediamine, $(CH_2O)_6N_2$,

which arises through the slow action of ammonia on the solid crystalline body. To prepare it I placed the latter between two watch glasses whose edges did not lie exactly over one another, and placed them, together with a small dish of dilute ammonia solution, under a glass bell. In this way the ammonia, which was renewed from time to time, had only limited access to the substance. After completion of the gas evolution that took place during the action, the lower glass was set over sulfuric acid to evaporate off the simultaneously formed liquid, and then the by-products — consisting mainly of hexamethylenamine and ammonium formate — were brought into solution by taking up in alcohol or water, and filtered off.

The yield of the new substance was only comparatively small, since the water appearing simultaneously seemed to work against its formation.

The substance crystallizes prismatically, in part completely cube-shaped.

Elemental analysis gave the following results:

1) 0.3414 g substance yielded 0.1832 g water and 0.4318 g carbon dioxide, corresponding to 5.96 pct. hydrogen and 34.50 pct. carbon.
2) 0.3238 g substance yielded 0.1759 g water and 0.4120 g carbon dioxide, corresponding to 6.03 pct. hydrogen and 34.70 pct. carbon.
3) 0.4875 g substance yielded 60.5 cc nitrogen at 736.5 mm, 21.5° C., corresponding to 13.61 pct. nitrogen.
4) 0.3246 g substance yielded 40.2 cc nitrogen at 735.0 mm, 19.8° C., corresponding to 13.68 pct. nitrogen.
5) 0.2608 g substance required, for neutralization of the ammonia after combustion with soda lime, 22.65 cc sulfuric acid [100 cc = 0.4090 g sulfuric anhydride], corresponding to 13.03 pct. nitrogen.

From these values the composition $C_6H_{12}O_6N_2 = (CH_2O)_6N_2$ can be derived, since this requires 34.61 pct. carbon, 5.77 pct. hydrogen, and 13.46 pct. nitrogen.

On slow heating the substance develops, without appreciable sublimation, an unpleasant, cheese-like odor; on stronger heating it deflagrates very vigorously, with hydrocyanic acid (prussic acid) appearing as an essential product; the latter also forms with deflagration on the action of nitrous acid.

The substance dissolves only with difficulty in water, alcohol, ether, ammonia, alkalis; in dilute acids it dissolves with decomposition, excellently on warming. In this process, with elimination of nitrogen in the form of ammonia, the starting substance is first re-formed, recognizable by its characteristic reactions, for:

on addition of excess ammonia, hexamethylenamine and ammonium formate form with evolution of oxygen;

on addition of alkali, formate salt forms with evolution of hydrogen; ammoniacal lead solution causes formation of lead peroxide; from potassium iodide solution, iodine separates out.

On further warming of the acidic liquid, the backward-formed starting substance decomposes into its components: formic acid, methyl aldehyde, and the ammonium salt of the acid used, while hydrogen simultaneously escapes.

With this substance I carried out the following quantitative determinations.

I. 1) On treatment with potassium iodide in [sulfuric-]acidic liquid, the substance decomposes gradually with intermittent shaking; 0.2208 g of it separated out, after 2 days, a quantity of iodine titrated with 51.4 cc sodium thiosulfate solution [100 cc = 0.0974 g oxygen]; this corresponds to 22.68 pct. oxygen.

2) In the same way were determined, from 0.1752 g substance after 2 days' action 22.82 pct., and from 0.1666 g after 3 days 22.77 pct. oxygen.

For the compound $(CH_2O)_6N_2$, however, the values obtained correspond to 3 atoms of oxygen, since 3 atoms of oxygen calculate to = 23.07 pct.

II. 1) 0.1011 g substance gave, after careful evaporation with excess hydrochloric acid and drying, 0.0558 g ammonium chloride, and 2) 0.1034 g substance gave 0.0562 g ammonium chloride; the nitrogen of the substance calculated as sal ammoniac amounts to 0.0520 g for 1) and 0.0532 g for 2)[3].

III. 0.3520 g substance was subjected, on addition of dilute sulfuric acid, to repeated distillation; in the distillate, formic acid and methyl aldehyde were first detected, and thereupon the ammonia formed was distilled off on addition of lye. Found 0.0564 g, calculated 0.0575 g ammonia.

From the reactions of the substance and the determinations carried out, I have derived the following decomposition equation:

$$(CH_2O)_6N_2 + 3\ H_2O + 2\ HCl = (CH_2O)_6O_3 + 2\ H_4NCl.$$

Since in this process the starting substance is demonstrably re-formed, the latter must then necessarily have the composition $(CH_2O)_6O_3$. But if one compares this with the formula $C_2H_6O_4$ mentioned at the outset, it follows that

$$3\ C_2H_6O_4 = (CH_2O)_6O_3 + 3\ aq.$$

The separation of iodine from potassium iodide in Experiment I therefore did not take place through the hexaoxymethylenediamine directly, but through the oxygen of the regenerated starting substance, the chiefly interesting product of the slow combustion of ether, which for the sake of simplicity may be designated by the name

Hexaoxymethylene Peroxide (hexaoxymethylene hyperoxide),
$(CH_2O)_6O_3 + 3$ aq,

a view which will find further confirmation through the following determinations.

The physical properties of the substance were already emphasized in the earlier reports.

Of its many and varied reactions I have investigated more closely the decompositions with alkalis and ammonia, proceeding as follows:

Into a eudiometer of about 500 cc capacity filled with mercury, which had a ground-glass stopper and a capillary tube at the upper end, I drew in a weighed solution of the substance by suction and then added standardized caustic soda in the same way. After the gas evolution had ended, the gas volume was read off over a column of water and mercury, and a portion of it was driven over into a combustion eudiometer for closer examination. After emptying the eudiometer, back-titration of the unconsumed lye was carried out, as well as determination of the methyl aldehyde as hexamethylenamine by standardized ammonia solution[4], and of the formic acid, after repeated distillation, both by titration and by means of mercuric chloride.

This procedure was followed because the quantities of decomposition products fluctuated, even when entirely uniform conditions were maintained during decomposition.

For example, 1 g substance required 20.2 cc normal caustic soda for decomposition, developed 185 cc hydrogen at 0° and 760 mm, yielded 0.0190 g CH_2O and (0.9068 g / 0.8892 g) CH_2O_2 determined by neutralization after distillation with $HgCl_2$; hence calculated carbon from CH_2O and CH_2O_2 = 0.80 + 23.6 = 24.4 pct.

For another experiment the values obtained from 1 g substance were: 17.9 cc normal caustic soda, 156 cc hydrogen, 0.0828 g CH_2O, (0.8030 g / 0.7776 g) CH_2O_2; and carbon = 3.30 + 20.95 = 24.3 pct.

The calculated carbon amounts thus agreed approximately with the carbon content of the substance = 25.53 pct. and confirm the correctness of the procedure applied[5]. At the same time it could be established from these and other experiments that the formic acid formation proceeded together with the hydrogen evolution, and that the quantities of both came out the greater the faster the decomposition took place and the more concentrated the lye used was.

All these facts led to the setting up of the following formula:

$$(CH_2O)_6O_3 \cdot 3\ aq + 6\ H_2O = 6\ HCO_2H + 6\ H_2O + 6\ H,$$

that is, the substance acts decomposingly in alkaline liquid on the water molecule; the oxygen of the latter joins the methyl aldehyde core and forms formic acid, which is thereby converted into the alkali salt, while the hydrogen escapes.

As is evident from the determinations cited, the conversion according to this equation does not take place completely quantitatively exactly: on the one hand, methyl aldehyde appears as a by-product, now in greater, now in lesser amount; on the other hand, the ratio of the decomposition lye to the hydrogen volume points to a partially occurring internal oxidation.

In acidic solution too, decomposition of water takes place through the substance, particularly on warming the solution; this is why the liquid prepared in the slow combustion of ether also continuously develops some hydrogen, particularly at rough surfaces or irregularities of the vessels.

On treatment with dilute sulfuric acid, 1 g substance yielded, after distillation, 0.2100 g CH_2O and 0.5980 g CH_2O_2; hence carbon = 8.4 + 15.6 = 24.0 pct.

Hydrogen peroxide, owing to its higher oxygen content, cannot bring about analogous decompositions of water. With regard to the release of the actively effective oxygen [= 17.0 pct.], the substance behaves more consistently than hydrogen peroxide, as is seen for example in the preparation of perchromic acid, in the reduction of potassium ferricyanide to Prussian blue, in the presence of ferric chloride, in the red coloration of titanic acid, etc. These reactions succeed only slowly in the acidified solution of the substance, but immediately if the decomposition has previously been initiated with ammonia.

With regard to the property of taking up nitrogen from ammonia with formation of water, the substance behaves like methyl aldehyde, which in this process forms hexamethylenamine; namely:

$$(CH_2O)_6O_3 \cdot 3\ aq + 2\ H_3N = (CH_2O)_6N_2 + 6\ H_2O.$$

Depending on concentration, the ammonia acts on the substance in yet another way: in gaseous ammonia it deflagrates immediately, with partial formation of yellow, resin-like products[6]; dilute ammonia, e.g. normal-strength liquid, causes evolution of oxygen, with mainly hexamethylenamine and ammonium formate being formed.

For example, 1 g substance yielded 46 cc oxygen, 0.3793 g CH_2O as $(CH_2)_6N_4$, and 0.2610 g CH_2O_2, with 15.17 + 6.81 = 22.0 pct. carbon.

Further, 1 g substance with concentrated ammonia, with slight evolution of oxygen, yielded 0.4631 g CH_2O, 0.1638 g CH_2O_2, with 22.8 pct. carbon.

Approximately, the decomposition with dilute ammonia can be expressed by the equation:

$$5[(CH_2O)_6O_3 \cdot 3\ aq] + 20\ H_3N =$$
$$(CH_2O)_6N_2 + 3\ (CH_2)_6N_4 + 6\ HCO_2NH_4 + 36\ H_2O + 6\ O.$$

With the help of this expression I calculated, on the basis of the preceding determinations, the crystal-water content of the substance, and obtained for this approximate values fluctuating between 16–23 pct.

The separation of hexamethylenamine can be accomplished in the following way: the filtrate remaining in the preparation of the hexaoxymethylenediamine is, after driving off the alcohol, evaporated on addition of baryta water at gentle temperature; the excess baryta is bound by carbon dioxide, and the $(CH_2)_6N_4$ is extracted from the mass, after prior drying, by repeated treatment with alcohol or chloroform. In the residue there then remains formate and some glycolate of baryta.

Of other modes of decomposition, I finally examined more closely the one with potassium iodide in acidic solution, and found that of the 3 atoms of the actively effective oxygen only 2–3 atoms enter into interaction; a part of it seems here too to be used for internal oxidation. The substance $(CH_2O)_6O_3$ regenerated from the hexaoxymethylenediamine by acids, on the other hand, acts in statu nascendi, as already shown, quantitatively exactly with 3 atoms.

I was unfortunately not in a position to confirm, by vapor-density determinations eventually to be carried out, the correctness of the molecular sizes of the two bodies; the evident relationships, however, which they bear to trioxymethylene or to hexamethylenamine respectively, do not make the assumption of more highly atomized complexes necessary.

A highly sensitive reaction, which serves for the detection of small amounts of hexamethylenamine and which I have often applied in the course of this work, is the copious separation of an orange-yellow body that forms with bromine water[7]. The latter loses bromine on standing in air, and there remains a sulfur-yellow, prismatically crystallizing, constant body which contains the amount of bromine corresponding to a dibromide; an addition product of the composition

$(CH_2)_6N_4Br_2$, a hexamethylenamine dibromide,

contains 53.33 pct. bromine. After decomposition of this body with sodium carbonate, evaporation, and ignition, there calculated from the separated silver bromide 1) 53.3 pct., 2) 53.5 pct. bromine. For the detection of methyl aldehyde by means of bromine water, ammonia was of course evaporated off beforehand.

To obtain a picture of the oxidation process of ether in slow combustion, I used the closed apparatus sketched alongside, whose arrangement appears understandable without further explanation. The 1 m long, cemented-in[8] platinum spiral was heated with the help of a battery. The entry of the ether vapor took place at *a*, the exit of the combustion products at *b*, insofar as such did not flow off into the vessel *c* fitted below. To this apparatus were attached, in the direction of the air current led through, suitable absorption devices.

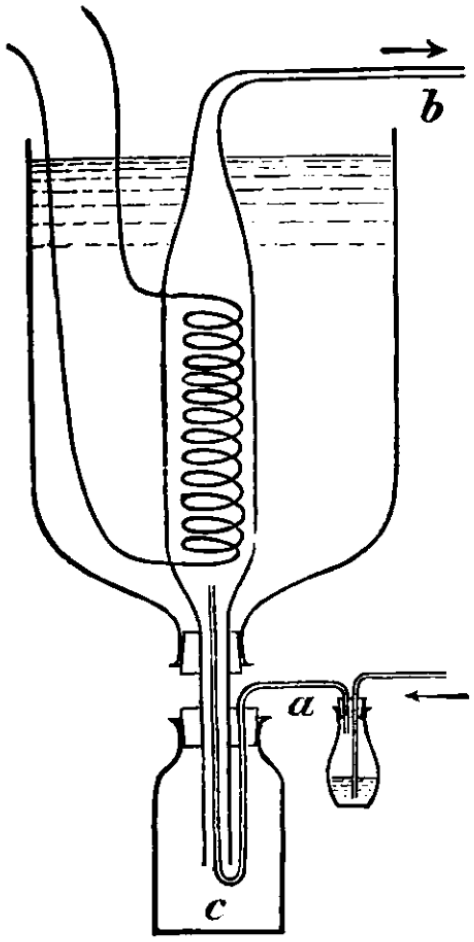


Fig. 2. Closed apparatus: ether vapor enters at a, combustion products exit at b, with vessel c below.

The products chiefly formed were carbon dioxide, water, acetaldehyde [secondarily acetic acid, methyl aldehyde, formic acid]: owing to the small platinum surface, the yield of finished hexaoxymethylene peroxide was only about 1.2 pct., of already decomposed material about 4.1 pct.

The more favorable yield was given by the preparation procedure mentioned at the outset using platinum sheet; there could be collected about 4.3 pct. finished substance and about 4.0 pct. of its decomposition products.

Following on from this work, I have also carried out some experiments on the action of ozone on ethylene gas and ether; for the former I obtained, with mist formation, chiefly the decomposition products of this peroxide, formic acid and methyl aldehyde; for the latter likewise, in small amounts, an oxidizing substance which, in my opinion, is identical with the product of the slow combustion of ether. According to statements by Berthelot, ethyl hyperoxide = $C_8H_{20}O_3$ forms here[9)].

Schönbein, in his numerous works on the actions of oxygen on organic matter, frequently mentions the occurrence of hydrogen peroxide in ether[10)]; likewise it has been observed there by v. Babo[11)] and Kingzett[12)].

In all the ether samples available to me, which through spontaneous decomposition contained an oxidizing substance, I did not succeed in demonstrating this with complete certainty as hydrogen peroxide; the reactions pointing to it could at least always be partly traced back to the ether product in question, and the attempts to prepare hydrogen peroxide from it in any way also remained unsuccessful.

Schönbein says in one of his last works[13]: “In view of the complexity of the chemical processes that take place in the slow combustion of ether, and of the so varied compounds appearing thereby, one of which, as mentioned, changes under one’s hands, a sufficient explanation of this process with all its accompanying circumstances seems to me one of the most difficult tasks that chemistry still has to solve, etc.”

Perhaps I may hope, through the foregoing contribution, to have come somewhat closer to the solution of this task.

To Privy Councillor Dr. Fleck, who permitted this work in the laboratory of the Chemical Central Office, I hereby express my most cordial thanks.

Dresden, December 18, 1885.

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1) These *Berichte* XIV, 602; *Ann. Chem. Pharm.* 217, 381.

2) Sitzungsber. d. Ges. Isis, Dresden 1882, 81.

3) This ammonium chloride contained small amounts of trimethylamine; with alkalis a partly faint herring-brine-like odor arose.

4) These *Berichte* XVII, 1333.

5) By-products appearing in subordinate amounts were: 1) an acid sharing the properties of glycolic acid, to be regarded as adhering to the substance under investigation — acetic acid did not belong to its decomposition products, as was earlier erroneously assumed; 2) a volatile decomposition product, smelling similar to ethylene, not examined more closely, which partly passed over into the hydrogen; the latter contained small amounts of carbon dioxide after deflagration.

6) The substance also deflagrates, with appearance of fire, on being brought together with platinum black.

7) Iodine likewise causes separation.

8) Brown, fat-like products that deposited on the spiral during the combustion process gradually rendered it inactive; a re-ignition (burning-out) for each new experiment was therefore necessary.

9) Chem. Centralbl. 1881, 534.

10) Journ. f. pr. Chem. 1866, 258, 282; 1868, 232.

11) Jahresber. d. Fortschr. d. Chem. 1864, 135.

12) These *Berichte* XIV, 248.

13) Journ. f. pr. Chem. 1868, 238, 239.

*English translation of: L. Legler, Berichte der Deutschen Chemischen Gesellschaft **18** (1885), No. 655, pp. 3343–3351. Figures reproduced from the original.*